

Project Build & Run Guide

This guide explains how to format, build, and run the project across platforms and also showcases the current rendering output and threading layout used in the path tracer.

Code Formatting

```
clang-format.exe -i src/*.h src/*.cpp
```

Windows: Build

```
set VCPKG_ROOT="$HOME/vcpkg"
vcpkg install
vcpkg integrate install
cmake -S . -B build -DCMAKE_TOOLCHAIN_FILE="${VCPKG_ROOT}/scripts/buildsystems/vcpkg.cmake"
cmake --build build
```

macOS: Build

```
brew install vcpkg
brew install pkg-config
git clone https://github.com/microsoft/vcpkg "$HOME/vcpkg"
export VCPKG_ROOT="$HOME/vcpkg"
vcpkg install
vcpkg integrate install
cmake -S . -B build -DCMAKE_TOOLCHAIN_FILE="${VCPKG_ROOT}/scripts/buildsystems/vcpkg.cmake"
cmake --build build
```

Debug Build

```
cmake -S . -B build -DCMAKE_TOOLCHAIN_FILE="${VCPKG_ROOT}/scripts/buildsystems/vcpkg.cmake"
cmake --build build
```

Release Build

```
cmake -S . -B build -DCMAKE_TOOLCHAIN_FILE="${VCPKG_ROOT}/scripts/buildsystems/vcpkg.cmake"
cmake --build build
```

Run (after a successful build)

```
cd build
./helloworld
```

Sample Output

Sphere layout overview: - Ground plane: large diffuse sphere at (0, -1000, 0).
- Ring of diffuse spheres: 12 objects of radius 0.62, resting at $y = r_{obj}$ on a circle of radius R. - Overhead emissive ring: 12 light spheres of radius 0.5 at $y = 6$, each emitting colored light.

```
// Ring scene layout:
const int N = 12;
const float R = 2.55f; // ring radius, gap between spheres
const float rObj = 0.62f;
const float objectY = rObj; // diffuse spheres rest on the ground plane
const float lightY = 6.0f;
const float rLight = 0.5f;
const float emission = 30.0f;
```

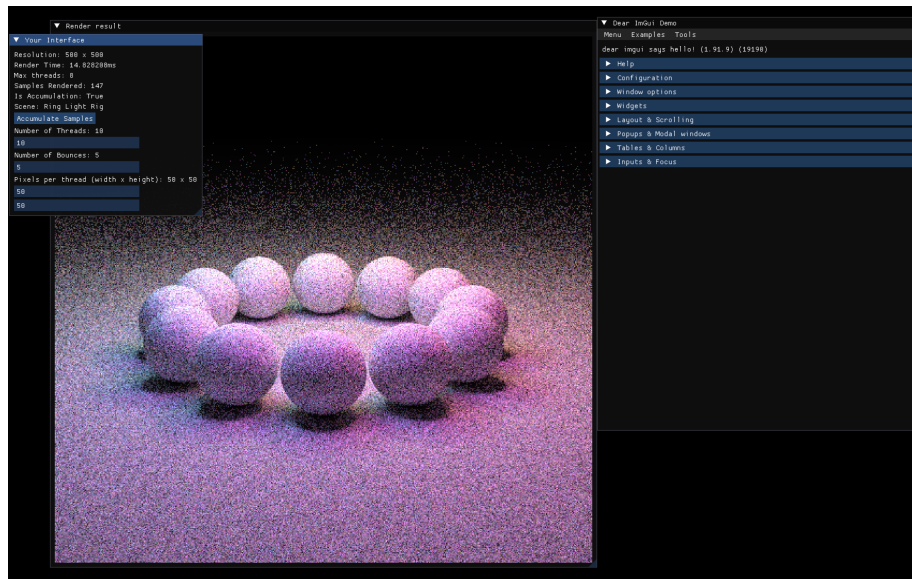


Figure 1: Sample render of the ring scene

Parallel Tiling Overview

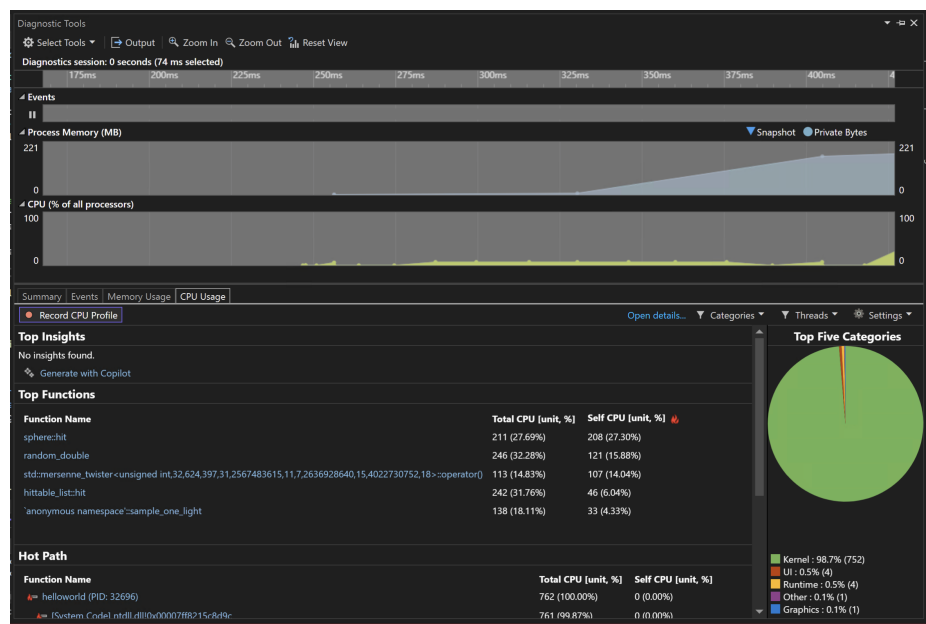


Figure 2: Thread tiling visualization